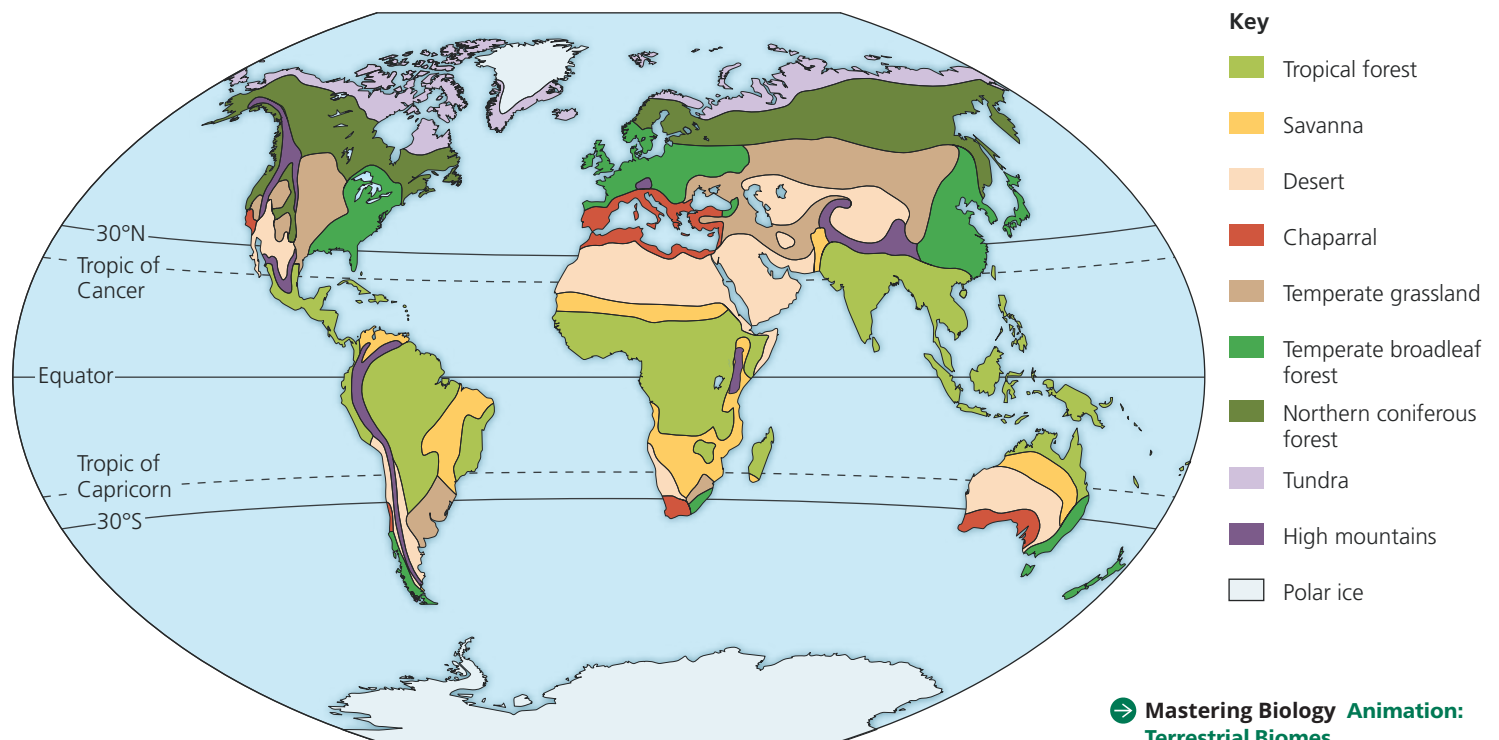


▼ **Figure 52.10** The distribution of major terrestrial biomes.



CONCEPT 52.2

The distribution of terrestrial biomes is controlled by climate and disturbance

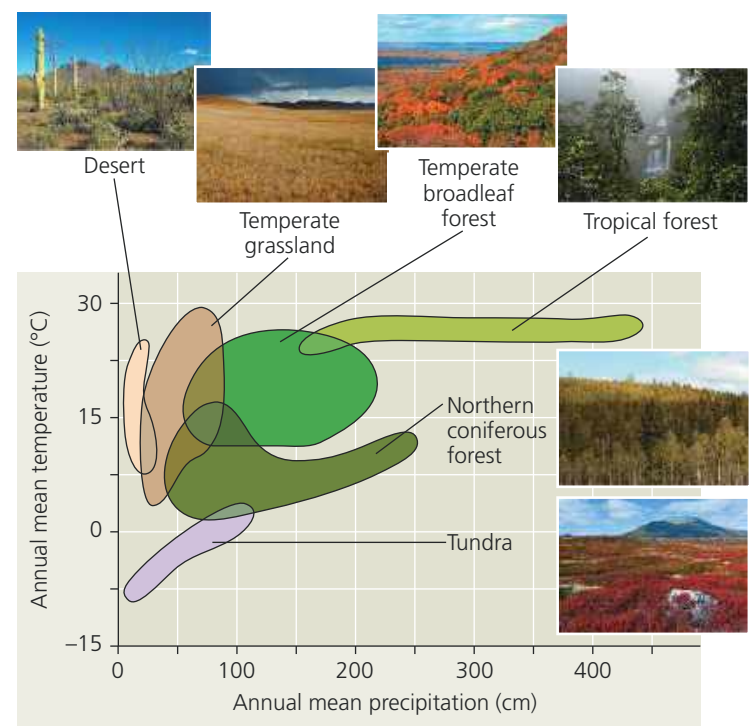
Earth's life is distributed on a grand scale in **biomes**, major life zones characterized by vegetation type in terrestrial biomes (or by the physical environment in aquatic biomes). On land, what determines where biomes are located?

Climate and Terrestrial Biomes

Because climate has a strong influence on the distribution of plant species, it is a major factor in determining the locations of terrestrial biomes (**Figure 52.10**). The importance of climate on the distribution of biomes can be shown in a **climograph**, a plot of the annual mean temperature and precipitation in a particular region. **Figure 52.11** is a climograph for some North American biomes. Notice, for instance, that the range of precipitation in northern coniferous and temperate forests is similar but that temperate forests are generally warmer. Grasslands are typically drier than either kind of forest, and deserts are drier still.

Factors other than mean temperature and precipitation also play a role in determining where biomes exist. Some areas in North America with a particular combination of temperature and precipitation support a temperate broadleaf forest, but other areas with similar values support a coniferous forest (see the overlap in **Figure 52.11**). One reason for

▼ **Figure 52.11** A climograph for some major biomes in North America. The areas plotted here encompass the ranges of annual mean temperature and precipitation in the biomes.



INTERPRET THE DATA Some arctic tundra ecosystems receive as little rainfall as deserts but have much more dense vegetation. What climatic factor could cause this difference? Explain.